Boss Blimp

Encounter design document

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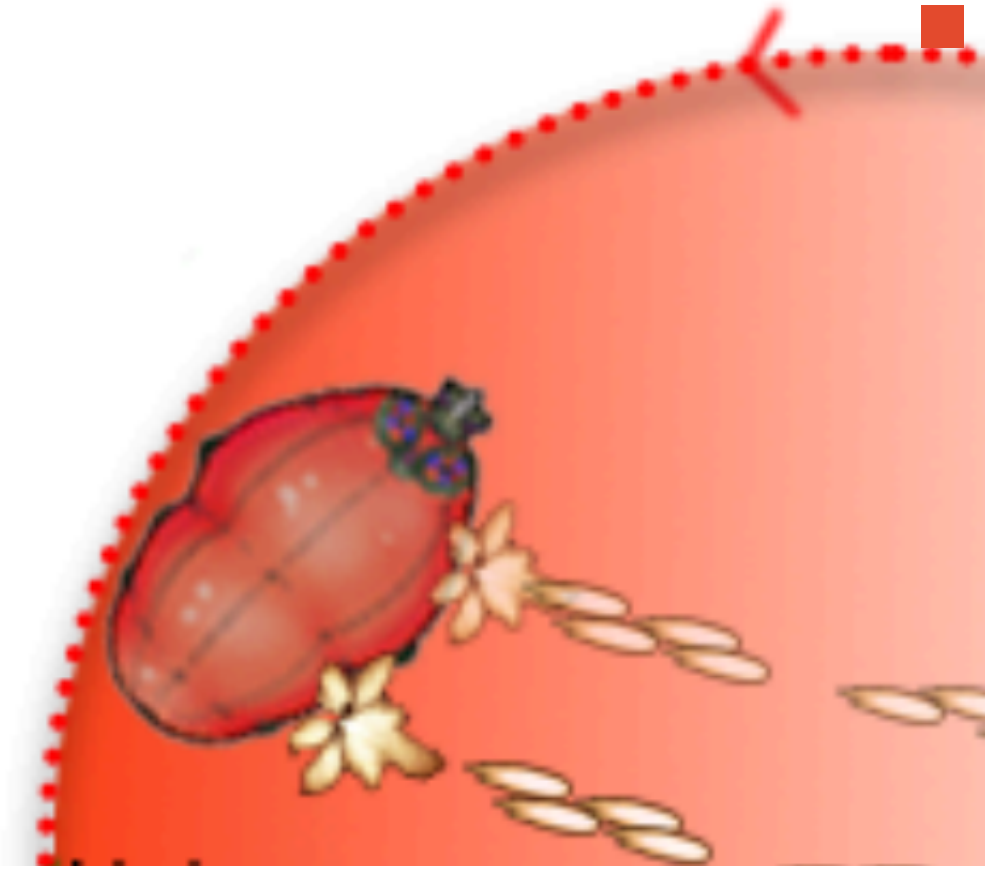

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Boss Blimp

Intro

The boss blimp is an encounter that requires the player to utilize their throwing, movement and grab controls. This boss is intended to mimic the behavior of the regular Blimp enemy, though with enhanced gun defenses and a modified attack pattern.

The Boss Blimp moves into the vicinity of the player and make an initial bombing run on them. After bombing the player, the blimp will use a figure-eight attack the player and drop bombs in the crossover. If the Blimp takes damage or loses its shield it will retreat to the 'Recovery' pattern.

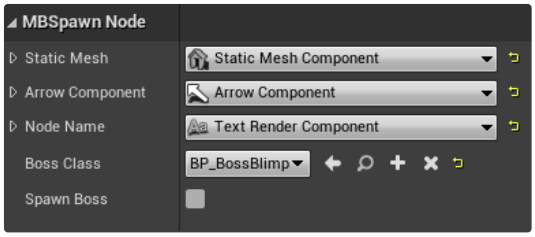
The boss blimp is not grabbable and is too heavy for damage by slapping.

Type

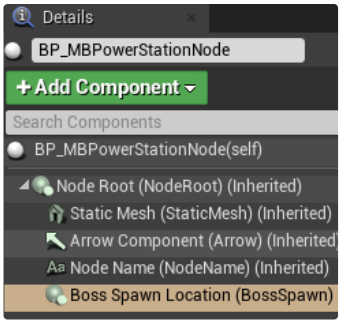
Airborne Blimp, Bomber.

Blimp Spawn

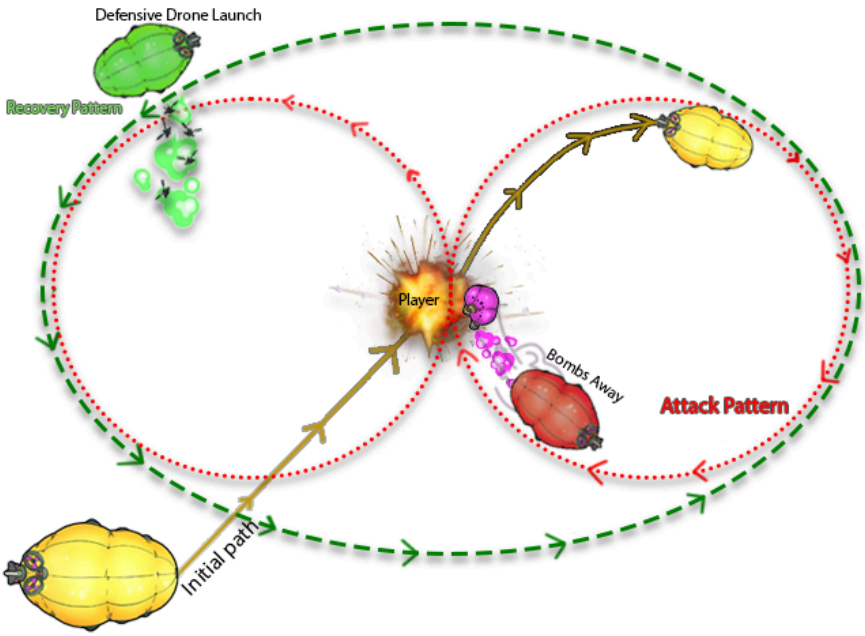
The Boss Blimp Spawns when key infrastructure is destroyed by the player.
The Designer may choose which structures spawn a blimp boss by checking the 'Spawn Boss' box on the corresponding generation node.
Bosses can be spawned from Shield Generators and Power Plants on destruction and Power Stations.



The Boss Blimp will spawn at a location placed by the Level Designer.



Movement



The blimp has two patterns.

In it's main 'Attack' posture it moves in a figure-eight over the player, dropping bombs as it passes over the player.

In it's defensive posture it moves faster and uses gunfire (bullets) to deter the player.

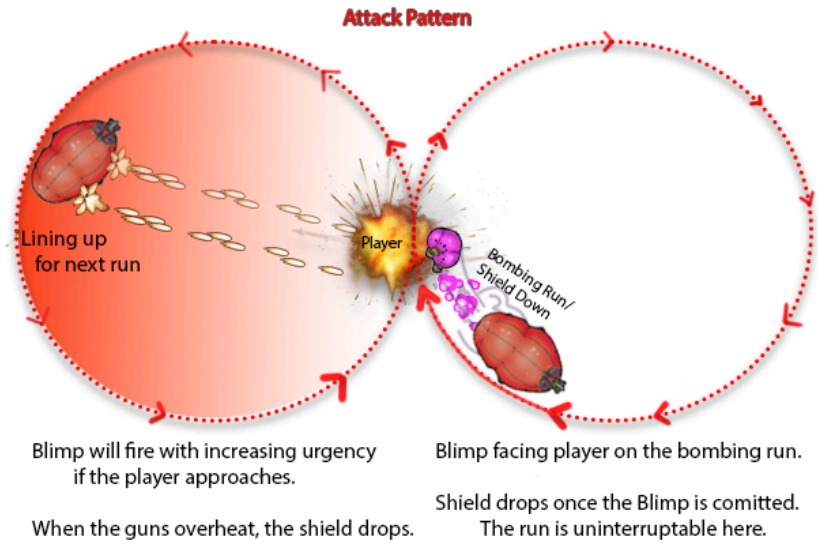
Variable	Explanation	Location
Ideal Height	The ideal height above ground level at which the boss attempts to fly.	Blimp Bp

Base Acceleration	The rate at which the blimp accelerates.	blimp BP
Acceleration Rate	How quickly the blimp accellerates, multiplier.	blimp BP
Cruising Acceleration	The speed at which the boss travel when they are not fleeing.	blimp BP
Top Acceleration	The rate at which blimps move when inside the attack range and when fleeing.	blimp BP
Maximum Firing Range	The upward limit of the blimps gun range	blimp BP
Threatened Distance	The distance at which the blimp begins to accelerate and shoot at the player.	blimp BP
Punch Damage Threshold	How hard the boss needs to be punched to take damage.	blimp BP
Bombing Run Accelleration	Affects the rate of acceleration when the boss is on the bombing run.	blimp BP
DeflectionPower	How much energy to add to deflected objects that hit the shield	blimp BP

Rotation

The blimp will be rotated towards it's current heading and the next node on it's path.
It only faces the player on the bombing run. When the blimp is in it's defensive pattern it will attempt to point it's guns (broadside) at the player.

Attack



The blimp will attack the player in a figure-eight pattern.
The bombing run is signalled by the sound of the bomb-launcher being armed and the appearance of a bomb mounted below the blimp. The bomb has a timed fuse indicated by a blinking light.

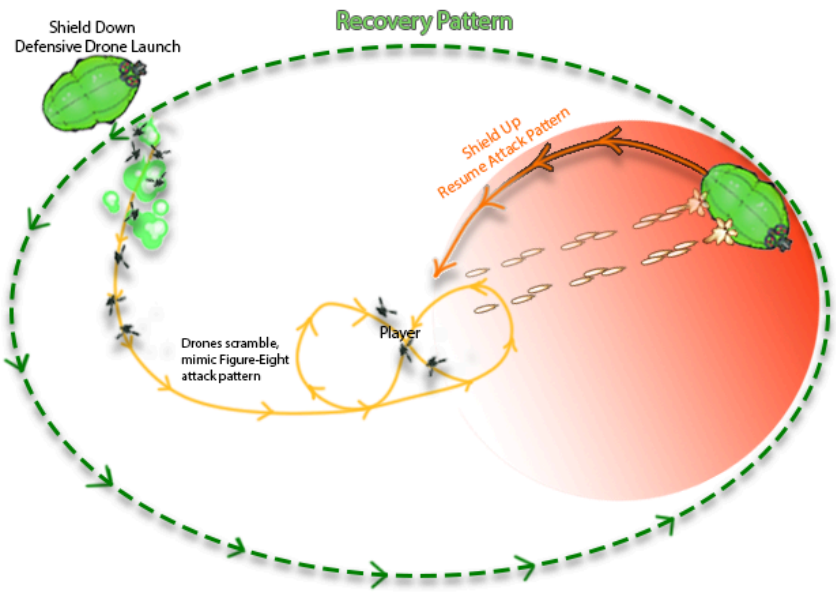
As the blimp aligns with the player and initiates the bombing run, it accelerates and drops the shield to loose the bomb. The bomb drops toward the player, bouncing if it collides with the ground. As the fuse nears it's end the flashes quicken. The player can pick this bomb up and throw it with the timer intact, or crush it to stop the timer.

After the bombing run, the blimp raises it's shields as it hits the outer edge of it's path. If the player advances on the Blimp at this stage it will begin defensively shooting bullets at the player. The guns will fire with increasing haste as the player approaches and they will begin to glow.
If the guns overheat the shield will drop until the bombing run is ended.

Variable	Explanation	
Shield Drop Distance	The distance from the center of the figure-eight attack pattern the boss must be for the shield to drop during the bombing run.	
Number of Laser Shots Per Burst	The initial number of shots in the laser burst.	
Starting Burst Cooldown Time	The time between bursts fired.	
Base Gun Fire Rate	Dictates the RPM of the laser gun before it begins to overheat	
Gun Heat Per Shot	Tracks how much heat the gun takes on each shot.	
Current Gun Heat	Tracks the current heat of the gun - should be left at 0.	
Gun Cooldown Value	How many points the heat will decrease by every Cooldown Rate cycle.	
Gun Cooldown Rate	Dictates how often the gun will cool by the cooldown in seconds.	
Number of Bombs to Drop	Dictates how many bombs the boss will	

	drop in it's undamaged state.	
Bomb Increase Per Hit	Dictates the increase in how many bombs the boss will drop each time the boss takes damage.	
Bomb Drop Interval	The amount of time between bombs dropped on a run.	

Defence



Blimp maintains defensive posture for one rotation of the player after taking damage.

Ignore 'Drone launch'

The Blimp is surrounded by an energy shield.

If the player touches the energy shield, the player takes damage and will be electrocuted (😫 Status Effects) making the hand numb and useless so the player has to shake it to recover control of that hand quickly (else the effects goes away after 8 seconds on its own, shaking it reduces the time to just 2 seconds). Upon touch the boss will act according to it's retreat sequence.

The shield is invulnerable against direct Special Power Damage but can be taken down with a bomb or multiple thrown objects.

If the Blimp's shield drops and it takes damage while the shield is lowered, the Blimp will divert into a defensive holding pattern while firing on the player.

Variable	Explanation	
Recovery laps	The number of rotations around the player that the blimp must do to recover.	
Movement Speed Increase Per Hit	the amount added to the boss's total speed when ever it takes a hit.	
Hits to Disable Shield	The number of thrown objects it takes to disable the shield	
Shield Heal Timer	Removes 1 hit to the shield each countdown.	
Shield Touch Damage	How much damage the shield deals to the player.	

Player strategy

1. Dodge incoming bombs by ducking and/or sidestepping.
2. Grab bombs off the ground and squeeze them, shutting the timer off and turning them into contact explosives.
3. Return the bomb to the sender by anticipating the bombing run and throwing the explosive as the Blimp's shield drops.

And/Or:

1. Attempt to disable the shield prematurely and land extra hits using rubble or the blimps own defensively deployed drones.

And/Or:

1. Attempt to short out the shield by baiting the Blimp into overheating it's guns while the player is using something as a shield.

The Blimps' shield redeploys and the boss attempts to resume it's Defensive Pattern.
The Blimp regenerates it's damaged weapon while inside the Shield and deploying drones. Once the weapons are regenerated it resumes the bombing run.

Staging

For each time the Blimp receives damage, the movement speed increases slightly, and it launches an additional bomb on its run and speeds up.

Once the player has successfully damaged the Blimp 3 times, explosions and particle effects go off the boss retreats into the horizon leaving a trail of smoke and sparks.

As with all player interaction, positive hit on Blimp should give the player clear feedback in the form of particle effects, sounds effects as well as animation (impulse from hit).

The boss will flash a darkened shade when damaged.

An undamaged Blimp emits only movement and attack particles.

A Blimp damaged once emits steam particles constantly as it leaks gas and occasional electrical arcs.

A Blimp damaged twice emits steam, electric arcs and is on fire.

Variable	Description	
Damage Speed Increase	How many units are added to the blimps speed when it takes damage. Applied evenly across all move speeds (but not to accelleration).	

Weapon configurations

Type	Ammo	Shots per burst	Pause between burst	Damage per shot
Gun Deck	Hitscan x2 per broadside	1-4	Distance Dependant	20
Larva Bomb	1 + 1 per damage taken	NA	Pattern Dependant	TBD

Parameters (OLD)

Variable	Function	Type / Value(s)
AttackSpeed	Determines the ‘charging’ time of the boss' attack in seconds	Int
DownTime	Determines the length in seconds of the vulnerable period after attack.	Int
RetreatValue	The distance (in units) at which the boss will decide the player is ‘too close’ and retreat	Float
RetreatDistance	The distance in units which the boss will attempt to cover when retreating.	Float
RetreatMoveSpeed	The speed at which the boss traverses the RetreatDistance.	Float
RetreatDelay	The delay between when the Boss decides to retreat and when it	Int

	executes the move in seconds.	
DefensiveCruisingSpeed	The speed at which the boss traverses its defensive holding pattern.	Float
TimeToRegenShields	The time in seconds that it takes for the shield to regenerate once downed.	Float
IdealAttackDistance	The distance in units from which the boss attempts to attack.	Float
MaxAttackDistance	The upper limit of the distance at which a boss can attack.	Float
AttackPower	The damage in HP that the boss' attack will inflict.	Int
AttackMoveSpeed	How fast the boss moves on its bombing run.	Float
TimeToLockAttack	The time interval in seconds after which an attack becomes unbreakable.	Int
ViableAttackDistance	The distance over which projectile attacks will still penetrate buildings.	Float
NormalMoveSpeed	The regular speed at which the boss moves	Float

	when not retreating or attacking.	
BossHealth	The number of ‘hits’ the boss takes to disable.	Int
DamageType	The type of damage this enemy deals.	Bullet, Laser, Etc.
GraceTime	The time interval in seconds for which the boss waits before attacking.	Int
AggroDistance	The detection distance in units for the payer.	Float
InitialBombsToDrop	The number of bombs to drop on a bombing run.	Int
RevengeBombMultiplier	The number of bombs to add to the run for each point of damage the boss absorbs	Int
TimeToOverheatGundeck	The amount of sustained fire (in seconds) it takes to overheat the gundeck	TBD